

Abhinav Gyawali

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Education

University of Texas at Austin PhD Computer Science Advisors Dixin Tang, Vijay Chidambaram	Aug 2026 (Expected) – TBD
Tribhuvan University BS Computer Science and Information Technology	Oct 2019 – Aug 2024

Publications

Mihai Budiu, Leonid Ryzhyk, Gerd Zellweger, Ben Pfaff, Lalith Suresh, Simon Kassing, Abhinav Gyawali , Matei Budiu, Tej Chajed, Frank McSherry, Val Tannen. DBSP: Automatic Incremental View Maintenance for Rich Query Language . <i>The International Journal on Very Large Data Bases (VLDB)</i>	May 2025
Abhinav Gyawali . SocketDB: DBMS with Data Streaming via WebSockets . <i>Deerwalk Journal of Computer Science and Information Technology (DJCSIT)</i>	Sep 2024

Experience

Feldera , Software Engineer	US, Remote Dec 2023 – Date
- Implemented Feldera support for sqlancer, an automated testing framework that generates and evaluates Feldera SQL queries, uncovering 31 previously undetected bugs. - Enhanced Feldera's SQL compiler with support for ARRAY and JSON (Variant) types, associated functions, type casting, nullability handling, and query optimizations. - Extended output connectors to enable writing real time query results to Postgres and Redis. - Designed S3-backed checkpoint synchronization (upload/download) and a "standby mode" to accelerate recovery from S3 checkpoints. - Designed and implemented the Feldera Python SDK, enabling programmatic control of pipelines, connector management, and integration with external systems. - Authored technical articles on using Feldera for OpenTelemetry Analysis and Accelerating Batch Processing.	
Invisid , Software Developer	Germany, Remote May 2022 – Nov 2023
Developed trackpad-rs, a specialized library to extract precise trackpad and magic mouse data from Apple's <i>MultitouchSupport</i> framework.	
Flow Webinar , Security Engineer (<i>Part time</i>)	US, Remote May 2020 – June 2021

Projects

DBSP	github feldera/feldera/dbsp
DBSP is an incremental computation engine where changes to the data set run in time proportional to the size of the change rather than the size of the data set.	

SocketDB

 [abhizer/socketdb](#)

- SocketDB is a lightweight SQL database that enables real-time updates through WebSockets.

Nyx-lang

 [abhizer/nyx-lang](#)

- Nyx-lang is a statically typed, tree-walking interpreted language with a type-checking mechanism to ensure safety and correctness.

Loogle-rs

 [abhizer/loogle-rs](#)

- Loogle is a "Local Google" like search engine based on Term Frequency - Inverse Document Frequency (TF-IDF).

Monkey-rs

 [abhizer/monkey-rs](#)

- Monkey-rs is my take on the concepts from the Writing an Interpreter in Go [book](#).

Awards & Certifications

UTCS Distinguished Graduate Fellowship	2026
SIGPLAN-M Mentee (Mentor: Professor Jonathan Balkind)	2025
DWIT Merit Based Scholarship	2021
DWIT Merit Based Scholarship	2020
AWS Educate Student Ambassador	2020
TedXMaitighar, Organizer	2019
Red Hat Certified Engineer (170-197-891)	2017
Red Hat Certified System Administrator (170-197-891)	2017